

# Bike Rebalancing in Jersey City: A Network Flow Approach to CitiBike Redistribution

## 1. Tentative Project Title

Bike Rebalancing in Jersey City: A Network Flow Approach to CitiBike Redistribution.

## 2. Team Members

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## 3. Problem Description and Data

Bike-sharing systems face a daily operational problem. By the end of each day, some stations have too many bikes while others are nearly empty. Overnight, the operator sends a small fleet of trucks to move bikes from surplus stations to deficit stations, so that every station is ready for the next morning. Our goal is to decide which stations each truck should visit and how many bikes to move, so that the total truck travel distance is minimized while station demand is met.

We model this as a minimum-cost network flow problem and solve it in four stages of increasing realism:

**Model 1 — Transportation LP.** Ignores trucks entirely: each surplus station ships bikes directly to deficit stations at minimum total bike-kilometers. Its solution is automatically integer because the constraint matrix is totally unimodular, and it serves as a lower bound for all later models.

**Model 2 — Truck routing MILP.** Adds realism: each truck has a fixed capacity, starts and ends at a depot, and follows a multi-stop route. Binary variables choose the route arcs; MTZ constraints eliminate subtours; load variables track how many bikes are on board along each arc. This is the core model of the project.

**Model 3 — Asymmetric road distances.** Same formulation as Model 2, but replaces straight-line (Haversine) distances with real driving distances from OpenStreetMap, which differ by direction due to one-way streets. We measure how much the optimal plan and cost change — i.e., whether the straight-line approximation is good enough in practice.

**Model 4 — Robust optimization.** Treats station demand as uncertain (the monthly average plus or minus a deviation) and uses the Bertsimas–Sim budget-of-uncertainty approach to build a plan that stays feasible even when up to  $\Gamma$  stations deviate from their averages. We test this plan out-of-sample on a second month of data.

**Data.** We use the public CitiBike trip data for Jersey City (May 2026 file, downloaded from <https://s3.amazonaws.com/tripdata/index.html>; about 20 MB unzipped). The file contains 95,350 trips. After removing 355 trips with missing end-station IDs and keeping only trips between the 108 Jersey City / Hoboken stations, we compute each station's net flow (arrivals minus departures). This yields 38 surplus stations and 64 deficit stations. Because total surplus and total deficit do not match exactly, we add a dummy node to balance the network. One month of data is sufficient for the main

models; we will also use a second month (June 2026) for out-of-sample validation. Distances between stations are computed from the GPS coordinates included in the data.

#### 4. Model Type and Approximate Size

The project uses linear programming, mixed-integer linear programming, and robust optimization. All models are implemented in Julia/JuMP and solved with the open-source solver HiGHS. Approximate sizes (with 108 stations plus one depot, and a fleet of 3 trucks):

| Model                     | Type | Variables (approx.)               | Constraints (approx.) |
|---------------------------|------|-----------------------------------|-----------------------|
| 1. Transportation problem | LP   | 2,400 continuous                  | 100                   |
| 2. Truck routing (MTZ)    | MILP | 36,000 binary + 37,000 cont./int. | 36,000 (mostly MTZ)   |
| 3. Asymmetric distances   | MILP | same as Model 2                   | same as Model 2       |
| 4. Robust counterpart     | MILP | Model 2 + ~200 continuous         | Model 2 + ~200        |

Model 1 solves in under a second. Model 2 is the largest model: it has about 12,000 binary routing variables per truck, and MTZ formulations are known to have weak LP relaxations. If the full 108-station instance is too slow to solve to optimality, our fallback plan is to (a) solve Models 2–4 on a 30–50 station sub-network chosen by ridership volume, or (b) sparsify the arc set by keeping only each station's nearest neighbors. Model 1 will still be solved on the full network in either case, so the estimates above represent an upper bound on the work required. All numbers are rough and may change as the project develops.